

Fig. 4: An actual-size, single-side PCB for HMI through EMG

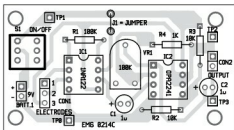


Fig. 5: Component layout for the PCB

or microcontrollers.

IC1 is specifically designed for single-supply operations, for which it includes voltage translators and over-voltage protection. The electrodes are directly attached to the muscle. Prior

to attaching the electrodes, clean the patient's skin with alcohol for removing dirt and apply an ultrasound gel. Two electrodes are attached over the muscle of interest and the third one (reference) is connected to an area preferably in the joints where there are less muscles. The reference electrode should only be connected at pin 3 of CON1. It is vital as it provides the common-mode signal.

The reference is connected to the virtual ground and pin 5 (Vref) of IC1, so care must be taken to isolate the patient from ground. The patient must not touch any grounded metal, which can cause drainage of common-mode voltage and thus raise the amplitude of the output and inject a lot of noise.

The output from instrumentation amplifier IC1 is fed to a non-inverting amplifier built around OPA2241 (IC2) via capacitor C1, which removes the DC offset from the electrodes. The amplified output signal from IC2 is sent to the PC oscilloscope via an audio jack to

get the display of the waveform.

Construction and testing

An actual-size, single-side PCB for the HMI circuit is shown in Fig. 4 and its component layout in Fig. 5. After assembling the circuit on PCB, enclose it in a suitable plastic case.

Fix a three-pin connector for the electrodes and two-pin connector for output. Also fix a two-pin connector for power supply on the front side of the panel. Please note that electrode connected to pin 3 of CON1 is the reference electrode and it should be in contact with a joint where there are less muscles. We used ECG electrodes during testing as they are easily available in medical stores. For troubleshooting, verify the voltages at different test points shown in the table. ●

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